Probabilities

→ Conditional probabilities and independence

Let (Ω, P) be a discrete probability space and let $A, B \subseteq \Omega$ with $P(B) > 0$. The conditional probability of A given B is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

// we know B occurs

→ eg:

Let $\Omega = \{1, \dots, 6\}$ and $P(A) = \frac{|A|}{6}$ for $A \subseteq \{1, \dots, 6\}$.

For $E = \{2, 4, 6\}$ and $S = \{1, 2, 3\}$,

$$P(E|S) = \frac{P(E \cap S)}{P(S)} = \frac{P(\{2\})}{P(S)} = \frac{1/6}{3/6} = \frac{1}{3}$$

→ Theorem 2.33

Let (Ω, P) be a discrete Probability space and let $B \subseteq \Omega$ with $P(B) > 0$. Then,

$$P(\cdot|B) : \mathcal{P}(\Omega) \ni A \mapsto P(A|B)$$

is a probability distribution on Ω.

Stochastics

→ Law of total probability

Let (Ω, P) be a discrete probability space and let $(B_i)_{i \in I}$ be an at most countable collection of disjoint subsets of Ω such that $\bigcup_{i \in I} B_i = \Omega$ and $P(B_i) > 0$ for $i \in I$.
Then for $A \subseteq \Omega$,

$$P(A) = \sum_{i \in I} P(A|B_i) P(B_i)$$

→ eg.

	Output	Defect rate
Machine 1	60%	5%
Machine 2	40%	10%

Consider a randomly selected part.
$A = \{\text{defect}\}$, $M_i = \{\text{machine } i\}$ $; i \in \{1,2\}$
We know

$$P(A|M_1) = 0,05 \quad ; P(A|M_2) = 0,1$$
$$P(M_1) = 0,6 \quad ; P(M_2) = 0,4$$

so that by Theorem 2.34

$$P(A) = P(A|M_1)P(M_1) + P(A|M_2)P(M_2) = 0,07$$

Stochastics

→ Bayes Formula

Let (Ω, P) be a discrete probability space and let $(B_i)_{i \in I}$ be an at most countable collection of disjoint subsets of Ω such that $\bigcup_{i \in I} B_i = \Omega$ and $P(B_i) > 0$ for $i \in I$. Then, for $A \subseteq \Omega$ with $P(A) > 0$,

$$P(B_j \mid A) = \frac{P(B_j)\, P(A \mid B_j)}{\sum_{j \in I} P(B_j)\, P(A \mid B_j)} \quad \text{for } j \in I$$

→ eg:

Consider a fire detector

$$A = \{\text{alarm}\}, \quad B = \{\text{fire}\}$$

Assume

$$P(B) = 10^{-5}, \quad P(A \mid B) = 0{,}99 \text{ and } P(A \mid B^c) = 10^{-4}$$

By 2.36,

$$P(B \mid A) = \frac{P(B)\, P(A \mid B)}{P(B)\, P(A \mid B) + P(B^c) P(A \mid B^c)} = 0{,}09$$

Stochastics

↳ Independence of events

Let (Ω, P) be a discrete probability space. Two events $A, B \subseteq \Omega$ are called independent if:

$$P(A \cap B) = P(A) \cdot P(B)$$

For $n \geq 2$, the events $A_1, \dots, A_n \subseteq \Omega$ are called independent if for all $I \subseteq \{1, \dots, n\}$ with $|I| \geq 2$,

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i)$$

For independent $A, B \subseteq \Omega$ with $P(B) > 0$,

$$P(A|B) = P(A)$$

↳ eg:

Let $\Omega = \{1, \dots, 6\}$ and $P(A) = \frac{|A|}{6}$ for $A \subseteq \{1, \dots, 6\}$ and let

$$A = \{2, 4, 6\}, B = \{1, 2\} \text{ and } C = \{2\}$$

Because of

$$P(A \cap B) = P(\{2\}) = \frac{1}{6} = P(A)P(B) = \frac{1}{6}$$

↳ A & B are independent

Since

$$P(A \cap C) = P(\{2\}) = \frac{1}{6} \neq P(A)P(C) = \frac{1}{12}$$

are not independent.

Stochastics – KVB – Ex 3

(1)

A mother has two children who are either males or female with the same probability. Specify a suitable probability space (Ω, P) and determine the probability that the other child is also a girl if you know that:

a) at least one children is a girl

$\Omega = \{0,1\}^2$ // 0 = female, 1 = male

$P(A) = \frac{|A|}{4}$

$A = \{$at least one children is a girl$\} = \{(0,1),(0,0),(1,0)\}$

$P(A) = \frac{3}{4}$

$B = \{$"the other child is also a girl"$\} = \{(0,0)\}$

$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{1/4}{3/4} = \frac{1}{3}$

b) the older child is a girl

$A = \{ \swarrow \} = \{(0,1),(0,0)\}$

$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{1/4}{2/4} = \frac{1}{2}$

Stochastics - KVB - Ex 3

(2)

A university has three faculties. 80% of women apply to Faculty 1, 15% to Faculty 2 and 5% to Faculty 3, 10% of men apply for F1, 20% for F2, 70% for F3.

Admission rate for women and men

	F1	F2	F3
women	60%	70%	80%
men	50%	60%	70%

Overall admission rate for woman & man.

Women:

$P(F1) = 80\%$ $\quad P(A_1|F1) = 60\%$

$P(F2) = 15\%$ $\quad P(A_1|F2) = 70\%$

$P(F3) = 5\%$ $\quad P(A_1|F3) = 80\%$

$$P(A_1) = P(A_1|F1)P(F1) + P(A_1|F2)P(F2) + \underbrace{P(A_1|F3)}_{P(F3)} = 0{,}625$$

Men:

$$P(A_2) = P(A_2|F1)P(F1) + P(A_2|F2)P(F2) + P(A_2|F3)P(F3) = 96$$

Stochastics – KVB – Ex 3

(3) Let A, B be events on some discrete probability space (Ω, P).

a) Let $P(B) > 0$. In which case is $P(A \cap B) = P(A|B)$ valid

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \overset{!}{=} P(A \cap B) \quad \rightsquigarrow P(B) = 1$$

b) Let $P(B) > 0$ and $P(A) > 0$. In which case is $P(A|B) = P(B|A)$?

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \overset{!}{=} \frac{P(A \cap B)}{P(A)} = P(B|A)$$

$$\rightsquigarrow P(B) = P(A) \quad \text{or} \quad P(A \cap B) = 0$$

c) In which case are the events independ.

$$P(A|B) = P(A)$$
$$P(A \cap B) = P(A) \cdot P(B)$$

d) $P(A \cap B) = 0 = P(\{\emptyset\})$

$$P(B|A) \cdot P(B) \overset{!}{=} 0$$

Stochastics - KVB - 6x3

(1)

prev = describes the relative proportion of people with the disease in the population under consideration

sens = indicates the proportion of people with the disease whose test is positive

spec = indicates the proportion of healthy people whose test is negative

$A = \{ill\}$ $B = \{\text{tested positive}\}$

sens = $P(B \mid A)$

spec = $P(B^c \mid A^c)$

a) sens = 1, spec = 0 $\Rightarrow$ all tests positive

sens = 0, spec = 1 $\Rightarrow$ all tests negative

Stochastik - KVB - Ex 3

(4)

5) Person ist getestet positiv - ist really ill

ill positiv

$P(A|B)$

Bayes: $P(B_i|A) = \frac{P(B_i)\,P(A|B_i)}{\sum_{j \in I} P(B_j)\,P(A|B_j)}$

ill Positiv

$$P(A|B) = \frac{\text{prev} \cdot \text{sens}}{\text{prev} \cdot \text{sens} + (1-\text{prev})(1-\text{spec})}$$

$\text{spec} = P(B^c|A^c)$

$P(B|A^c) + P(B^c|A^c) = 1$!

Stochastics - K1 / B-Ex3

(5)

Let (Σ, P) be a discrete probability space with $\Omega = \{a,b,c,d,e\}$

$P(\{a\}) = P(\{b\}) = P(\{e\}) = 1/4$

$P(\{c\}) = P(\{d\}) = 1/8$

$A = \{a, c, d\}$, $B = \{b, c, d\}$, $C = \{a, b\}$

a) Show that $A \& B$, $A \& C$, $B \& C$ are independent

$P(A) = \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = \frac{2}{4} = 1/2$

$P(B) = \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = \frac{2}{4} = 1/2$

$P(C) = \frac{1}{4} + \frac{1}{4} = \frac{2}{4}$

$P(A \cap B) = P(\{c,d\}) = \frac{1}{4} = P(A) \cdot P(B) = \frac{2}{4} \cdot \frac{2}{4} = \frac{1}{4}$

$P(A \cap C) = P(\{a\}) = \frac{1}{4} = P(A) \cdot P(C) = \frac{1}{4}$

$P(B \cap C) = P(\{b\}) = \frac{1}{4} = P(B) \cdot P(C) = \frac{1}{4}$

c) Show that $A \& B \& C$ not indep.

$P(A \cap B \cap C) = P(\{\emptyset\}) = 0 \neq P(A)P(B)P(C)$

Stochastics - KVB

Def (σ-algebra)

For a set $\Omega \neq \emptyset$, a family $\mathcal{F}$ of subsets of Ω is called a σ-algebra on Ω if

(i) $\Omega \in \mathcal{F}$

(ii) $A \subset \mathcal{F} \Rightarrow A^c \in \mathcal{F}$

(iii) $A_1, \dots, A_n \in \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

→ $\emptyset \in \mathcal{F}$

→ $\mathcal{F}$ is closed under finite unions

→ $\mathcal{F}$ is closed under at most countable intersections

eg:

For $\Omega \neq \emptyset$, $\mathcal{F}_1 = \{\emptyset, \Omega\}$ and $\mathcal{F}_2 = \mathcal{P}(\Omega)$ are σ-algebras.

Def probability space

A probability space $(\Omega, \mathcal{F}, P)$ consists of a set $\Omega \neq \emptyset$, a σ-algebra $\mathcal{F}$ on Ω and a function $P: \mathcal{F} \to [0,1]$ such that

(i) $P(\Omega) = 1$

(ii) for all disjoint $A_1, A_2, \dots \in \mathcal{F}$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Rg:

For a discrete probability space (Ω, P) and $\mathcal{F} = \mathcal{P}(\Omega)$, $(\Omega, \mathcal{F}, P)$ is a probability space

Stochastics - KVB

→ General Probability space

For $\Omega \neq \emptyset$ and $A \subseteq P(\Omega)$ let $\delta(A)$ be the smallest δ-algebra on Ω such that $A \subseteq \delta(A)$

→ eg:
For $\Omega = \{1,2,3,4\}$,
$$\delta = (\{\{1\}\}) = \{\emptyset, \{1\}, \{2,3,4\}, \{1,2,3,4\}\}$$

→ 2.5 3:

For $\Omega = \mathbb{R}$ and $A = \{(a,b] : a,b \in \mathbb{R}, a \leq b\}$,

$$\mathcal{B}(\mathbb{R}) = \delta(A)$$

is the Borell δ-algebra on $\mathbb{R}$. For $\Omega = \mathbb{R}^d$, $d \geq 2$,

$$A = \{\bigtimes_{i=1}^{d} (a_i, b_i] : a_i, b_i \in \mathbb{R}, a_i \leq b_i, i \in \{1,\dots,d\}\}$$

$$\mathcal{B}(\mathbb{R}^d) = \sigma(A)$$

is the Borell - δ-algebra on $\mathbb{R}^d$.

→ eg: The Lebesgue measure is the unique map

$$\lambda_d : \mathcal{B}(\mathbb{R}^d) \to [0,\infty]$$ such that

$$\lambda_d\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \lambda_d(A_i)$$

for disjoint $A_1, A_2, \dots \in \mathcal{B}(\mathbb{R}^d)$ and

$$\lambda_d\left(\bigtimes_{i=1}^{d} (a_i, b_i]\right) = \prod_{i=1}^{d} (b_i - a_i)$$

for $a_i, b_i \in \mathbb{R}$ with $a_i \leq b_i$ for $i \in \{1,\dots,d\}$

For $\Omega = \mathbb{R}$ and $P(\cdot) = \lambda_1(\cdot \cap (0,1])$,

$(\Omega, \mathcal{B}(\mathbb{R}), P)$ is a probability space

Stochastics - KVB

↝ Random Variable

Let (Ω, P) be a discrete probability space. A map $X : \Omega \to \mathbb{R}$ is called a random variable.

↝ eg:

Let $\Omega = \{0,1\}^2$ and $P(A) = \frac{|A|}{|\Omega|}$ for $A \subseteq \Omega$.

Then,

$$X : \Omega \to \mathbb{R}, \quad \omega \mapsto X(\omega) = \begin{cases} 2 & , \omega = (1,1) \\ 1 & , \omega = (1,0) \text{ or } \omega = (0,1) \\ 0 & , \omega = (0,0) \end{cases}$$

↝ 3.3:

For a discrete probability space (Ω, P) and a random variable $X : \Omega \to \mathbb{R}$ define $P_X : \mathcal{P}(X(\Omega)) \to [0,1]$ by

$$P_X(A) = P(X \in A) = P(X^{-1}(A)) = P(\{\omega \in \Omega : X(\omega) \in A\})$$

for $A \subseteq X(\Omega)$. Then, $(X(\Omega), P_X)$ is a discrete probability space.

(P_X: distribution of X)

↝ eg:

In Example 3.2, X has the distribution $P_X : \mathcal{P}(\{0,1,2\}) \to [0,1]$ with

$$P_X(\{0\}) = \frac{1}{4} \qquad P_X(\emptyset) = 0$$

$$P_X(\{1\}) = \frac{2}{4} < 1/2 \qquad P_X(\Omega) = 1$$

$$P_X(\{2\}) = 1/4$$

Stochastics I KIB

eg:
→ 3.6 (Indicator random variable):
For a discrete probability space (Ω, P) and $A \subseteq \Omega$ the indicator function of A

$$1_A : \Omega \ni \omega \mapsto 1_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}$$

is a random variable. Its distribution is given by

$$P(1_A = 1) = P(A) \text{ and } P(1_A = 0) = P(A^c) = 1 - P(A)$$

→ eg: 3.7 (Binomial distribution):
A random experiment with outcomes success / failure and success probability $p \in [0,1]$ is repeated n times. Let X be the number of successes. For $k \in \{0, \dots, n\}$, we have

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k},$$

X has a Binomial distribution with the parameters n and p.

→ eg: 3.8 (Poisson Distribution):
We say that a random variable Y follows a Poisson distribution with parameter $\lambda \in [0, \infty)$ if

$$P(Y = k) = \frac{\lambda^k}{k!} e^{-\lambda} \quad \text{for } k \in \mathbb{N}_0$$

→ eg: 3.10: (Hypergeometric distribution):
We consider an urn with b blue and r red balls, from which we draw $n \leq b + r$ balls without replacement. Let the random variable X describe the number of drawn blue balls.
For $k \in \{0, \dots, n\}$ we obtain

$$P(X = k) = \frac{\binom{b}{k}\binom{r}{n-k}}{\binom{b+r}{n}}$$

KVB-

Stochastik - Ex 4

(1)

Let $r, b, a, n \in \mathbb{N}_0$. In the Urn model there are, in the beginning, r red balls and b black balls in an urn. One randomly draws a ball from this urn and then puts it back together with a new balls of the same colour. This process is repeated n times. Show that the event of drawing a red ball from the urn in the n-th step has the same probability as drawing a red ball in the first step.

R_k = event of drawing a red ball in k'th step

B_k = event of drawing a black ball in k'th step

$P(R_1) = \frac{r}{r+b}$; $P(B_1) = \frac{b}{r+b}$; Show $P(R_n) = \frac{r}{r+b}$

Induction:

$n = 1$: $P(R_1) = \frac{r}{r+b}$

$n \mapsto n+1$: $P(R_{n+1} | R_1) = \frac{r+a}{r+a+b}$; $P(R_{n+1} | B_1) = \frac{r}{r+b+a}$

$\hookrightarrow P(R_{n+1}) = P(R_{n+1} | B_1) P(B_1) + P(R_{n+1} | R_1) P(R_1)$

$= \frac{r}{r+b}$ ∎

Stochastics - KVB - 6.4

(2)

In a system with the two possible characters dot and dash.
4% of the transmitted dots are received as dashes and
2% of the transmitted dashes as dots.
ratio of dots to dashes is 3/5

$\hookrightarrow \frac{P(Dot)}{P(Dot^c)} = \frac{3}{5}$ $\qquad 1 = P(Dot) + P(Dot^c)$

$P(Dot) = \frac{3}{5} P(Dot^c) \longrightarrow 1 = \frac{3}{5} P(Dot^c) + P(Dot^c)$

$1 = \frac{8}{5} P(Dot^c)$

$\Leftrightarrow P(Dot^c) = \frac{5}{8}$

$P(Dot) = \frac{3}{8}$

a) Determine the probabilities that a Dot was sent if a dot was received, or that a dash was sat if a dash was received

RecDot = receive dot ; Dot = send Dot

$P(RecDot^c \mid Dot) = 0,04$, $P(RecDot \mid Dot^c) = 0,02$

$$P(Dot \mid RecDot) = \frac{P(Dot)P(RecDot \mid Dot)}{P(Dot)P(RecDot \mid Dot) + P(Dot^c)P(RecDot \mid Dot^c)}$$

$1 - P(RecDot^c \mid Dot) = P(RecDot \mid Dot) = 0,96$

$$P(Dot \mid RecDot) = \frac{\frac{3}{8} \cdot 0,96}{\frac{3}{8} \cdot 0,96 + \frac{5}{8} \cdot 0,04} = 0,9664$$

Stochastic - KVB-Ex 4

(2)

b) $P(\text{Rec Dol}) = P(\text{Rec Dol} \mid \text{Dol}) P(\text{Dol})$
$+ P(\text{Rec Dol} \mid \text{Dol}^c) P(\text{Dol}^c)$

$= 0,96 \cdot \frac{3}{8} + 0,02 \cdot \frac{5}{8} = 0,3725$

$P(\text{Rec Dol}^c) = P(\text{Rec Dol}^c \mid \text{Dol}) P(\text{Dol})$
$+ P(\text{Rec Dol}^c \mid \text{Dol}^c) P(\text{Dol}^c) = 0,6275$

$\frac{P(\text{Rec Dol})}{P(\text{Rec Dol}^c)} = 0,5936$

Stochastics KVB – Ex. 4

(3)

Let $\Omega = \{1,2,3,4\}$. Determine the smallest possible σ-algebra $\mathcal{F}$ for which $\{2,3\} \in \mathcal{F}$ and $\{4\} \in \mathcal{F}$.

$$\mathcal{F} = \{ \emptyset, \underbrace{\{1,4\}}_{\{1\}}, \underbrace{\{2,3\}}_{\{2,3,4\}}, \{1,2,3\}, \{4\}, \{1,2,3,4\} \}$$

i) $\Omega \in \mathcal{F}$

ii) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$

iii) $A_1, A_2, \ldots \in \mathcal{F} \Rightarrow \bigcup_{i \in \mathbb{N}} A_i \in \mathcal{F}$

Statistics - KVB - Ex 4

(4)

The following table describe the distribution of the random variable X.

x	-2	-1	0	1	2
P(X=x)	0,1	0,2	0,1	0,2	0,4

Determine $P(X>0)$, $P(|X-1| \leq 1)$ & The distribution of $Y = X^2$

- $P(X>0) = P(\{1,2\}) = 0,6$
- $P(|X-1| \leq 1) = P(\{-1, 0, 2\}) = 0,1 + 0,2 + 0,4 = 0,7$
-

y	4	1	0
P(Y=y)	0,5	0,4	0,1

$P(X=0) = 0,1$

$P(X=1) + P(X=-1) = 0,4$

$P(X=2) + P(X=-2) = 0,5$

Stochastik - KVB - Ex 4

(5)

On a road, four traffic lights regulate traffic.
Each of them allows or prohibits a car to proceed with probability 0,5; whereby this happens independently of the other traffic lights.

a) Specify a suitable probability space and define a random variable X, which describes the number of traffic lights the car passes before it has to stop for the first time. (Or 4 if it doesn't)

$\Omega = \{0,1\}^4$ 0 - stop; 1 - pass

$P = \frac{|A|}{16}$

$X: \Omega \to \{0,1,2,3,4\}$

$\omega = (\omega_1, \omega_2, \omega_3, \omega_4) \mapsto \max\{k \in \{0,\dots,4\}:$

$\omega_i = 1 \text{ for } i \in \{0,\dots,k\}$

b)
$P(X=0) = P(\{\omega_1,\dots,\omega_4\}: \omega_1 = 0) = \frac{2^3}{16} = \frac{1}{2}$

$P(X=1) = P(\{\omega_1,\dots,\omega_4\}: \omega_2 = 0) = \frac{2^2}{16} = \frac{4}{16} = \frac{1}{4}$

$P(X=2) = P(\{\omega_1,\dots,\omega_4\}: \omega_3 = 0) = \frac{2}{16} = \frac{1}{8}$

$P(X=3) = P(\{\omega_1,\dots,\omega_4\}: \omega_4 = 0) = \frac{1}{16}$

$P(X=4) = P(\{\omega_1,\dots,\omega_4\}: \omega_i \neq 0 \; i \in \{1,\dots,4\}) = \frac{1}{16}$